Do Co-resident Grandmothers Lighten Mothers' Care Load? Three-Generation Households and Women's Time in India

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Abstract

Co-resident grandmothers are widely assumed to substitute for mothers in childcare. India, where most three-generation households are patrilocal and formal care for children under six is limited, is a demanding test of that assumption. The Time Use Survey 2024 records a 24-hour diary for every household member aged six and above, so I observe on the same day the time of mothers *and* of the grandmothers who live with them. I identify the generation above each married woman from roster relationship codes, including the common patrilocal case in which the mother is the head's daughter-in-law, and I keep out of the mother sample women whose co-resident young child is most likely a grandchild. Among 49,992 married mothers aged 18 to 49 in households with a child under six, 36% live with a grandmother. Conditional on the mother's and husband's characteristics, consumption, household composition, caste, religion, and state and day fixed effects, a co-resident grandmother is associated with 2.1 minutes (SE 1.3) less primary childcare by the mother a day and 0.4 minutes (SE 1.7) less paid work. Grandmothers themselves average 86 minutes of childcare a day, and in the older generation a young grandchild is associated with 79.5 minutes (SE 1.1) more childcare for grandmothers and 29.8 minutes (SE 0.7) more for grandfathers. In sum, grandmothers' presence is not associated with significantly less childcare by mothers, and children in three-generation households receive more childcare from adult members in total. Their presence is not associated with significantly more paid work by mothers. The results bear on how far family care can stand in for formal childcare.

Keywords: grandparents, childcare, three-generation households, patrilocality, time use, women's employment, India

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1 Introduction

The family is India's main childcare institution. Public provision for children under six runs mainly through the *anganwadi* centres of the Integrated Child Development Services, which focus on supplementary nutrition, health and pre-school education rather than full-day care, and women's labour force participation has long been low by international standards and slow to rise with education and income (Fletcher et al. 2017; Klasen and Pieters 2015; Chatterjee et al. 2018). In this setting a common presumption, in policy discussion and in the family itself, is that co-resident grandparents, above all grandmothers, absorb a large share of the care of young children and so free mothers to work, rest or invest in other children. The presumption has empirical support elsewhere. In the United States, Europe, China and Japan, the availability of grandparents is associated with, and in several quasi-experimental designs causes, higher maternal employment (Compton and Pollak 2014; Posadas and Vidal-Fernández 2013; Arpino et al. 2014; Bratti et al. 2018; Maurer-Fazio et al. 2011; Sasaki 2002; Ogawa and Ermisch 1996).

There are reasons to doubt that the same holds in India. Residence after marriage is predominantly patrilocal, so the co-resident grandmother is typically the mother-in-law, and the anthropological and economic literature describes the young daughter-in-law as the junior member of a hierarchy in which her mobility, work outside the home and access to services are shaped by her husband's parents (Dyson and Moore 1983; Kandiyoti 1988; Jayachandran 2015; Anukriti et al. 2020). Women in joint households in rural India are less likely to be employed than women in nuclear households (Dhanaraj and Mahambare 2019). A three-generation household is also a larger household, with more meals to cook and sometimes a frail elder to look after. Co-residence can therefore *add* to a mother's work at the same time as it brings a potential substitute for her childcare.

These two forces, which I call the *substitution* channel and the *constraint* channel, have opposite implications for the mother's time. Which dominates is an empirical question that most existing studies cannot answer directly. Surveys of labour supply record whether a grandparent lives nearby and whether the mother works, but not what either of them does with the day. Surveys of grandparents record whether they "look after grandchildren", but not the mother's time on the same day. As a result, the literature has mostly inferred the substitution of care from its labour-market consequences.

This paper measures the substitution directly. India's Time Use Survey 2024 (National Statistics Office 2025) records a 24-hour diary, coded to the international ICATUS 2016 classification (United Nations Statistics Division 2016), for every member aged six and above in each sampled household. Diaries of mothers and of the grandparents who live with them are thus kept for the same day. The household roster's relationship codes let me identify, for each married woman, the generation above her. For a woman who is the head or the head's spouse, that generation consists of the head's parents or parents-in-law. For a woman who is a married child of the head or a married child's spouse, it consists of the head and the head's spouse, which is the typical patrilocal arrangement

of a daughter-in-law living in her husband's parents' household. An identification that treats only "parent of head" as a grandparent misses this second case entirely. A further correction matters as much: in a household where a son and his wife live with his parents and a child under six is present, the head's wife is usually that child's grandmother, not its mother. I exclude such women from the mother sample (2,189 of the 52,181 married women aged 18 to 49 in households with a child under six) and study them instead as grandmothers.

The analysis has four parts. First, I compare the time use of mothers of children under six with and without a co-resident grandmother, conditional on a rich set of characteristics of the mother, her husband and the household, and on state and diary-day fixed effects. The outcomes cover the whole day: primary childcare split into active care and passive supervision, childcare as a secondary activity, domestic work, care of dependent adults, employment, free time and sleep. Second, I add the grandmother's own childcare to the mother's to ask whether three-generation households provide the same care more cheaply or more care in total. Third, I turn to the older generation itself and estimate how the presence of a young grandchild is associated with grandmothers' and grandfathers' childcare, domestic work, paid work and free time; this shows the cost of the substitution to the person who provides it. Fourth, I ask which grandparents matter (younger or older grandmothers, working or not, grandmothers or grandfathers), how the association varies across settings, and whether it vanishes where it should, among mothers whose youngest child is of school age.

The main finding is that a co-resident grandmother is not associated with a statistically distinguishable change in the mother's primary childcare: the estimate is -2.1 minutes (SE 1.3). The association with the mother's paid work, including travel, is -0.4 minutes (SE 1.7), with her domestic work 1.1 minutes (SE 1.7), and with her free time 4.3 minutes (SE 1.5). Grandmothers who live with a child under six report 85.9 minutes (SE 1.1) of childcare a day, 37% of the combined childcare of mother and grandmother. Summed over all household members aged 15 and above, primary childcare is 88.6 minutes (SE 2.3) higher in households with a grandmother, conditional on the number of children. Grandmothers' care therefore adds to what children receive without measurably replacing what mothers provide. On the older generation's side, a child under six in the household is associated with 79.5 minutes (SE 1.1) more childcare for grandmothers and 4.7 minutes (SE 2.5) more paid work, against 29.8 minutes (SE 0.7) more childcare for grandfathers.

The paper contributes to three literatures. For the economics of grandparental childcare, it replaces inferences from employment with direct, same-day time measures for both generations, and it does so in a patrilocal, low-income setting where the constraint channel is plausibly strongest. For the study of family structure and women's work in South Asia, it shows how a household's kinship structure maps into the minutes of each member's day, separating the care that a co-resident mother-in-law provides from the work her presence entails. For the literature on child penalties (Kleven, Landais, and Søgaard 2019; Kleven, Landais, Posch, et al. 2019; Angelov et al. 2016; Cortés and Pan 2023), it provides evidence on one margin, informal family care, along which the penalty could in principle be cushioned, and on how far that cushion extends in a setting without formal alternatives.

Methodologically, the paper documents relationship-based rules for assigning young children and grandparents in household rosters that do not record parent identifiers, rules that apply to many household surveys in South Asia.

The estimates are conditional associations, not causal effects. Living arrangements are chosen jointly with fertility, work and care, and the survey is a single cross-section. I address selection in four ways that do not rely on an instrument: by comparing arrangements that are selected for different reasons, by asking whether the associations follow the gradients that a substitution story predicts (in grandmother's age, employment and sex), by a placebo on mothers of school-age children, and by coefficient-stability bounds (Oster 2019). None of these is decisive alone; together they constrain the interpretation.

The rest of the paper is organised as follows. Section 2 reviews the literature. Section 3 sets out a simple framework that organises the two channels and the predictions I test. Section 4 describes the data and the identification of mothers, young children and grandparents. Section 5 presents the empirical strategy, Section 6 the results and Section 7 robustness checks. Section 8 discusses interpretation, policy and limitations, and Section 9 concludes.

2 Related literature

2.1 Grandparental childcare and maternal labour supply

A large literature finds that grandparents' availability raises mothers' labour supply. In the United States, Compton and Pollak (2014) show that married women who live close to their mothers or mothers-in-law have higher labour force participation, and they attribute the effect to the availability of childcare; Posadas and Vidal-Fernández (2013) find, with data from the National Longitudinal Survey of Youth, that grandparent-provided childcare raises mothers' labour force participation. In Europe, Dimova and Wolff (2011) show with data from the Survey of Health, Ageing and Retirement in Europe that downward time transfers, mainly grandchild care, raise maternal labour supply; Arpino et al. (2014) find a positive effect of grandparental support on mothers' participation in Italy using an instrumental-variable approach; and Aassve et al. (2012) compare the relationship across European countries with the Generations and Gender Survey. Bratti et al. (2018) exploit Italian pension reforms that raised retirement ages, and so reduced grandmothers' availability, and find that the reforms lowered the participation of their daughters. Earlier, Del Boca (2002) had shown that the availability of childcare and of part-time jobs shapes Italian women's joint decisions about participation and fertility. Aparicio-Fenoll and Vidal-Fernández (2015) find that grandmothers' own labour force participation is associated with lower fertility among their daughters, consistent with grandmothers' time being a binding input into the care of young children. García-Morán and Kuehn (2017) argue that grandparent-provided care, though cheap, comes with constraints on mothers' labour market choices.

The evidence from Asia is closer to the Indian setting because co-residence, not proximity, is the

usual arrangement. Ogawa and Ermisch (1996) find that co-residence with parents raised Japanese married women's full-time employment, and Sasaki (2002) confirms a positive effect of co-residence with parents or parents-in-law on Japanese married women's participation after treating living arrangements as endogenous. In urban China, Maurer-Fazio et al. (2011) show that co-residence with parents or parents-in-law was associated with higher participation of married women between 1982 and 2000, while young children lowered it, and Shen et al. (2016) study co-residence with elderly parents and the labour supply of women in China, where co-resident parents can be sources of care as well as recipients of it. Descriptive studies document how widespread grandparental care is in Europe (Hank and Buber 2009), China and Korea (Chen et al. 2011; Ko and Hank 2014). Igel and Szydlik (2011) show that the welfare state shapes the form of that care: grandparental childcare is more common but less intensive where public childcare is more developed.

Two features of this literature motivate the present paper. First, almost all of it measures the *consequence* of grandparental care (maternal employment) rather than the care itself. Second, it comes overwhelmingly from settings in which co-residence or proximity is not tied to a strong patrilocal hierarchy. Where the grandmother is the mother-in-law and the household's head, her presence may bring authority over the daughter-in-law's time as well as help with it.

2.2 The older generation's time and intergenerational transfers

Grandparental childcare is a transfer of time between generations. Economic models of intergenerational transfers distinguish altruistic from exchange motives (Cox 1987; Altonji et al. 1997), and Konrad et al. (2002) show how adult children's location choices respond strategically to the expectation of caring for parents. Cardia and Ng (2003) show in a quantitative model that grandparents' time transfers are an important input into childcare and matter for how mothers' labour supply responds to childcare subsidies. Albertini et al. (2007) document that European regimes differ in the frequency and intensity of time and money transfers across generations. Time transfers have an opportunity cost for the giver. Rupert and Zanella (2018) find that the arrival of grandchildren lowers grandmothers' labour supply in the United States, and Lumsdaine and Vermeer (2015) find that a new grandchild raises the probability that women retire. The evolutionary literature on the "grandmother hypothesis" and on kin effects on child survival makes the same point from a different angle: grandmothers' investment in grandchildren is a central and costly part of human life histories (Hawkes et al. 1998; Sear and Mace 2008). Diaries for the older generation let me measure this cost directly, alongside the benefit to the mother.

2.3 Family structure, patrilocality and women's work in India

India differs from the settings above in ways that could reverse the sign of the association between co-residence and mothers' work. Dyson and Moore (1983) argued that the kinship system of northern India, with village exogamy and patrilocal residence, limits women's autonomy relative to the south; Rahman and Rao (2004) revisit the thesis with newer data, and Jejeebhoy and Sathar (2001)

show that women's autonomy in India and Pakistan varies more with region than with religion. Kandiyoti (1988) describes the "patriarchal bargain" in which young brides accept subordination in their husband's household in exchange for later authority as mothers-in-law. Economic studies document the consequences. Anukriti et al. (2020) show that co-resident mothers-in-law restrict daughters-in-law's mobility and social networks in rural Uttar Pradesh, which lowers their use of family planning, and Debnath (2015) relates co-residence with in-laws to lower female autonomy. Status concerns can lead households to withdraw women from work outside the home (Eswaran et al. 2013), and social norms more broadly are a barrier to women's employment (Jayachandran 2021; Field et al. 2021). Dhanaraj and Mahambare (2019) find that women in nuclear households in rural India are more likely to work than women in joint households, and Afridi et al. (2018) decompose the decline in rural married women's participation over two decades and highlight the role of changes in household characteristics such as education and income. Explanations for low and stagnant participation also emphasise the structure of economic growth and labour demand (Klasen and Pieters 2015; Lahoti and Swaminathan 2016; Chatterjee et al. 2018).

These studies observe employment, autonomy or decision-making, not time. Whether grandmothers do childcare that mothers would otherwise do, and what the mother does with any time released, cannot be answered without diaries for both women on the same day.

2.4 Childcare, child penalties and policy

The child penalty literature shows that the arrival of children accounts for a large and persistent share of the gender gap in earnings in high-income countries (Kleven, Landais, and Søgaard 2019; Kleven, Landais, Posch, et al. 2019; Angelov et al. 2016; Cortés and Pan 2023), and Goldin (2014) argues that the remaining gap reflects the cost of temporal flexibility. Formal childcare is the standard policy response. Universal childcare raised maternal employment in Quebec (Baker et al. 2008; Lefebvre and Merrigan 2008), and public kindergarten raised it in the United States (Gelbach 2002). In Norway, by contrast, the expansion of subsidised childcare mainly crowded out informal arrangements and had little effect on maternal employment (Havnes and Mogstad 2011), a reminder that the effect of formal care depends on what it replaces. In a low-income setting, Clark et al. (2019) find that subsidised daycare raised poor mothers' employment in Nairobi. Where formal care is scarce, the question is how much of the work that formal care would do is already done by family members, and at what cost to whom. The answer bears both on the size of the child penalty in India and on what an expansion of full-day childcare could achieve.

2.5 Measuring childcare in time diaries

Time-diary research shows that childcare is the most gendered of household tasks and that its measurement is sensitive to definitions (Sayer et al. 2004; Bianchi et al. 2006; Craig and Mullan 2011; Hook 2006). Guryan et al. (2008) document a strong education gradient in parents' time with children, and Kalenkoski et al. (2005) and Connelly and Kimmel (2009) model parental childcare as

a household allocation. Primary childcare understates the time parents are responsible for children, because much supervision is recorded as a secondary activity or not recorded at all (Folbre et al. 2005; Folbre 2006; Kalenkoski and Foster 2008; Craig 2007). I report active and passive primary care separately and childcare recorded as a secondary activity, because a grandmother may take over supervision while the mother keeps the care that requires her, or the reverse.

3 A framework: substitution and constraint

To organise the empirical analysis, consider a household in which a mother of a young child allocates her day among childcare c_m, domestic work h_m, paid work e_m and leisure l_m, with $c_m + h_m + e_m + l_m = T$. The child requires care $C = g(c_m, \theta c_o)$, where c_o is the childcare of an older-generation member and θ captures how far her care can stand in for the mother's. Care tasks differ in θ: supervision is easily shared, while breastfeeding, feeding and some medical or educational care are less so. The household's domestic workload is $H = H_0 + \kappa N$, increasing in the number of members N, and it is divided between the mother and other adults. Paid work earns a return $w(1 - \tau)$, where τ is a wedge, from norms or from the authority of senior members, that lowers the value the household places on the mother's work outside the home.

A co-resident grandmother changes three things. She can supply c_o, which lowers the care the mother must provide (the *substitution* channel); the size of this effect rises with θ and with the grandmother's capacity, which declines with age and poor health and falls when she works herself. She raises N and hence H, and she may need care herself (a *workload* channel). And in a patrilocal household she may raise τ for a daughter-in-law (a *norms* channel). I group the last two as the constraint channel. The framework yields the following predictions, each of which I take to the data.

1. If substitution is present, mothers' childcare is lower with a grandmother, and more so for passive care (high θ) than for active care.
2. Whether released time reaches paid work depends on τ and on the workload channel. If the constraint channel is strong, released childcare time shows up in domestic work or leisure, not employment.
3. The grandmother's own time should show the other side: more childcare and, if her time is scarce, less paid work, domestic work or leisure when a young grandchild is present.
4. Substitution should be larger with grandmothers who are younger (a proxy for health and stamina) and not employed. Under prevailing gender norms it should be smaller with grandfathers than with grandmothers.
5. Substitution of childcare should be concentrated where the youngest child is under six. Among mothers whose youngest child is of school age, the grandmother association with childcare should be smaller. The constraint channel does not depend on the child's age, so associations with domestic work should persist.

The framework is deliberately simple. It does not model the choice of living arrangement, which is

the main threat to interpretation and which I discuss in Section 5.

4 Data and measurement

4.1 The Time Use Survey 2024

The Time Use Survey (TUS) 2024 was conducted by the National Statistics Office between January and December 2024 (National Statistics Office 2025). It covers a nationally representative sample of households in rural and urban India, drawn in a stratified multi-stage design with First Stage Units (villages and urban blocks) as primary sampling units. Every member aged six and above reports a 24-hour diary for the day before the interview, from 4 a.m. to 4 a.m., in time slots that can contain several activities. Each activity is coded to ICATUS 2016 and flagged as major or minor (secondary). I use every activity record: 10,211,478 records for 533,719 diarists in 139,487 households and 9,969 First Stage Units. Minutes of major activities are summed to the person-day, so each diary sums to 1,440 minutes (100.0% of diaries do so exactly). All estimates use the survey multipliers, and standard errors are clustered on the First Stage Unit (Abadie et al. 2023). Appendix Table 9 reports the construction checks.

4.2 Identifying mothers and children under six

Children under six do not keep diaries, and the roster does not record the identifier of each child's mother. I recover the number of under-six members as household size minus the number of diarists (method: direct count (file contains members under 6)), which relies on every member aged six or above keeping a diary. The share of households in which this residual is negative, an indicator of a violation, is –.

The candidate mothers are married women aged 18 to 49 whose relationship to the head is head, spouse of head (head level), married child, or spouse of a married child (junior level). Women with other relationships (for example a brother's wife, coded "other relative") are excluded because the roster does not identify the generation above them. Assigning a young child to a mother requires a rule. In a household in which a married child of the head lives with a spouse, a child under six is far more likely to be the junior couple's child, that is, the head's grandchild, than a late-born child of the head. I therefore exclude from the mother sample head-level women who live with a junior-level married woman aged 15 to 49. This removes 2,189 women. Without this step, the head's wife in a patrilocal household would be counted as a "mother without a co-resident grandmother", although she is the grandmother; the comparison would then contrast grandmothers with mothers. The remaining ambiguity concerns households with more than one junior couple, where the young child could belong to either; a robustness check restricts the sample to households with a single married woman of childbearing age (86% of the weighted sample).

The analysis sample consists of the 49,992 remaining women in households with at least one child under six. Of these, 33% are at junior level, and among junior-level mothers 95% are daughters-in-law

of the head (the remainder are married daughters living in their parents' household). The husband is linked to 83% of mothers through the roster. Husbands who live elsewhere, a common situation in states with high male out-migration, are not linked; I keep these mothers in the sample with an explicit indicator rather than dropping them, as an analysis that required husband controls would do implicitly.

4.3 Identifying grandparents

For a head-level mother, the generation above her consists of members coded "father, mother, father-in-law or mother-in-law of head" (relationship code 7). If the mother is the head's wife, a woman in this group is usually her mother-in-law; if the mother is herself the head, it is her own mother or her husband's mother. For a junior-level mother, the generation above consists of the head and the head's spouse aged 40 or above: her parents-in-law if she is the head's daughter-in-law, her own parents if she is the head's daughter. A co-resident grandmother is at least one woman in this generation, and a co-resident grandfather at least one man. Because I build the link by matching each mother to the specific members of the generation above her, the grandmother is never the mother herself (0 self-links), and the count of linked grandmothers agrees with the panel's count of older-generation women for 99.8% of mothers. Each mother is then linked to her grandmothers' and grandfathers' own diaries.

By this definition, 36% of mothers of children under six live with a grandmother and 28% with a grandfather; 4.5% live with a grandfather but no grandmother. Among mothers with a co-resident grandmother, 83% are at junior level, so most three-generation households in the sample are of the patrilocal type in which the young family lives in the older generation's household. The median age of the youngest co-resident grandmother is 57 years, 60% of grandmothers are under 60, and 30% report employment as their usual principal activity status.

4.4 Time outcomes

Childcare is ICATUS 411–419 plus accompanying own children (442). Active care (411–415) covers physical care, medical care, teaching, talking and reading, and playing; passive care (416) is minding children. I also measure childcare recorded as a secondary (minor) activity, unpaid domestic services (division 3) and within it food and meals (311–319), the largest domestic task and the one most directly increased by household size, care of dependent adults (421–429, 443), employment including related travel (division 1), free time (divisions 7 and 8) and sleep (911–919). Employment status is the usual principal activity status. Every category is defined once, on the three-digit code, and the categories that make up the day partition the ICATUS codes exactly.

Table 1 compares the raw time use of mothers with and without a co-resident grandmother. Mothers without a grandmother report 146.1 minutes (SE 0.9) of primary childcare a day and mothers with one 144.8 minutes (SE 1.1). The raw comparison mixes the presence of the grandmother with everything that differs between joint and nuclear households, including region, caste, consumption,

Table 1: Mothers of children under six: minutes per day by grandmother co-residence

Outcome	No grandmother	Grandmother present
Childcare (primary)	146.1 (0.9)	144.8 (1.1)
Active childcare	141.2 (0.9)	139.8 (1.1)
Passive childcare (minding)	3.6 (0.2)	3.5 (0.2)
Childcare as secondary activity	45.6 (1.0)	33.6 (0.8)
Unpaid domestic work	358.7 (1.2)	340.9 (1.6)
of which food and meals	253.8 (1.1)	242.0 (1.3)
Care of dependent adults	0.1 (0.0)	0.4 (0.1)
Employment (incl. travel)	48.1 (1.2)	49.7 (1.4)
Free time	182.6 (1.1)	193.5 (1.4)
Sleep	527.9 (1.0)	530.6 (1.0)
Employed (usual status, share)	0.2 (0.0)	0.2 (0.0)

Notes: Married women aged 18–49 at head or junior level in households with a child under six, excluding head-level women who live with a junior married woman of childbearing age. Survey-weighted means; FSU-clustered standard errors in parentheses. Employed is a share.

the mother's age and the number of children, which motivates the regression approach.

5 Empirical strategy

For mother i in household h, state s and diary day d, I estimate

$$y_{ih} = \beta\, GM_h + X'_{ih}\gamma + \mu_s + \tau_d + \varepsilon_{ih},$$

where y_{ih} is a time outcome in minutes per day (or employment status) and GM_h indicates a co-resident grandmother. X_{ih} includes the mother's age band and education, the husband's education and usual employment (with a separate category for mothers without a linked husband), the household's per-capita consumption quintile, urban residence, the number of children under six and aged 6 to 14, the number of the mother's own children aged 6 to 17 (with an indicator where these cannot be attributed), caste group, religion, whether any member needs special care, and whether the diary was reported by a proxy. State fixed effects absorb regional kinship systems and labour markets, and day-of-week fixed effects absorb the weekly cycle. Regressions are weighted by the survey multipliers (Solon et al. 2015) and standard errors are clustered on the First Stage Unit.

I deliberately do not control for household size in the main specification. A grandmother's presence raises household size mechanically, and the workload that comes with a larger household is part of the constraint channel the paper aims to measure. A robustness check adds household size and the number of other adult women.

The main threat to interpretation is selection into co-residence. Three-generation households differ from nuclear ones in wealth, caste, region and norms, and young mothers in patrilocal joint households may differ in unobserved preferences, in their bargaining position and in the age and

health of their children. Co-residence can also respond to care needs: a grandmother may move in because a grandchild is born or because she herself needs care. I therefore read β as a conditional association and assess how sensitive it is in four ways.

First, I compare arrangements selected for different reasons. Junior-level mothers live in the household of the older generation, the default patrilocal arrangement; head-level mothers whose parents or in-laws live with them have taken the older generation in, which is more often driven by the grandparents' need for support. Similar estimates across arrangements would make a single selection story less plausible. As a matter of data, almost every junior-level mother lives with a grandmother (only 2,054 do not, compared with 15,196 who do), because a widowed father-in-law heading the household is rare. Where the cells are too small (fewer than 30 mothers on either side) I do not report the junior-level estimate, and identification within that group must come from the head-level comparison.

Second, I test the gradients that the substitution channel predicts but selection on fixed household traits does not obviously predict: larger associations with younger grandmothers, with grandmothers who are not employed, and with grandmothers rather than grandfathers. Third, I estimate the same specification for mothers whose youngest child is aged 6 to 14 and who have no child under six in the household. Substitution of childcare should be weaker for these mothers; selection on persistent household traits would generate similar associations in both groups. Fourth, I report the bias-adjusted coefficient of Oster (2019), which uses the movement of β and of R^2 when controls are added to bound the role of unobservables under the assumption of proportional selection (see also Altonji et al. 2005).

The older generation's trade-offs are estimated on a separate sample: members aged 45 to 80 in the generation above a married woman of childbearing age (the head and spouse in households with a junior couple, and parents or in-laws of the head in households with a head-level couple). For these 26,062 women and 19,382 men, I regress the same time outcomes on an indicator for a child under six in the household, with controls for age band, education, marital status, urban residence, consumption quintile, children aged 6 to 14, caste, religion, special care needs and proxy reporting, and state and day fixed effects. The comparison is between older-generation members whose grandchildren are young and those whose grandchildren are older or not yet born, which depends on the timing of the junior couple's fertility rather than on the older generation's decision to co-reside.

6 Results

6.1 Mothers' time

Table 2 reports the main estimates. A co-resident grandmother is associated with 2.1 minutes (SE 1.3) less primary childcare by the mother a day, significant only at the 10% level. Relative to the mean of mothers without a grandmother, 146.1 minutes, this is 1% of the baseline. The split by type

Table 2: Co-resident grandmother and mothers' time use

Outcome	Mean	Coefficient	SE	N
Childcare (primary)	146.1	-2.09*	1.27	49,992
Active childcare	141.2	-2.17*	1.24	49,992
Passive childcare (minding)	3.6	-0.05	0.26	49,992
Childcare as secondary activity	45.6	-9.60***	1.09	49,992
Unpaid domestic work	358.7	1.14	1.70	49,992
of which food and meals	253.8	5.01***	1.36	49,992
Care of dependent adults	0.1	-0.08	0.07	49,992
Employment (incl. travel)	48.1	-0.41	1.73	49,992
Free time	182.6	4.34***	1.51	49,992
Sleep	527.9	-3.10***	1.12	49,992
Employed (usual status, share)	0.18	-0.01***	0.00	49,992

Notes: Each row is a separate regression of the outcome (minutes per day, or a 0/1 indicator for usual employment) on an indicator for a co-resident grandmother and the full control set, with state and day-of-week fixed effects. Mean: outcome mean for mothers without a co-resident grandmother. Survey-weighted; FSU-clustered standard errors. * p<0.10, ** p<0.05, *** p<0.01.

of care shows -2.2 minutes (SE 1.2) for active care and -0.0 minutes (SE 0.3) for passive minding. Neither component differs significantly with a grandmother's presence. Childcare recorded as a secondary activity changes by -9.6 minutes (SE 1.1).

The remaining rows show where the mother's day goes. Unpaid domestic work differs by 1.1 minutes (SE 1.7), of which food and meals by 5.0 minutes (SE 1.4). Domestic work does not differ significantly, so the larger household does not translate into a measurably larger domestic load for the mother. Care of dependent adults differs by -0.1 minutes (SE 0.1). Employment including travel differs by -0.4 minutes (SE 1.7), and the probability that the mother's usual status is employed is 1.4 percentage points (SE 0.5) lower. Neither the minutes of paid work nor the employment rate differ significantly, so any time grandmothers free up does not reach the labour market. Free time differs by 4.3 minutes (SE 1.5) and sleep by -3.1 minutes (SE 1.1).

Taken together, the evidence for the substitution channel in mothers' own time is weak, and there is no evidence that co-residence with a grandmother is associated with more paid work by mothers.

Table 3: Childcare minutes per day by provider, households with a child under six

Outcome	No grandmother	Grandmother present
Mother	146.1 (0.9)	144.8 (1.1)
Husband (if linked)	44.3 (0.6)	24.4 (0.4)
Grandmother(s)	0.0 (0.0)	85.9 (1.1)
Grandfather(s)	3.2 (0.2)	18.0 (0.5)
Other members aged 15+	7.8 (0.3)	51.3 (1.4)
All members aged 15+	201.4 (1.3)	324.4 (2.7)

Notes: Unit: mothers in the analysis sample. Minutes of primary childcare of each household member, survey-weighted means with FSU-clustered standard errors. Other members include husbands who are not linked through the roster. All members is the household total and includes care given to other children in the household.

6.2 Who provides childcare in three-generation households

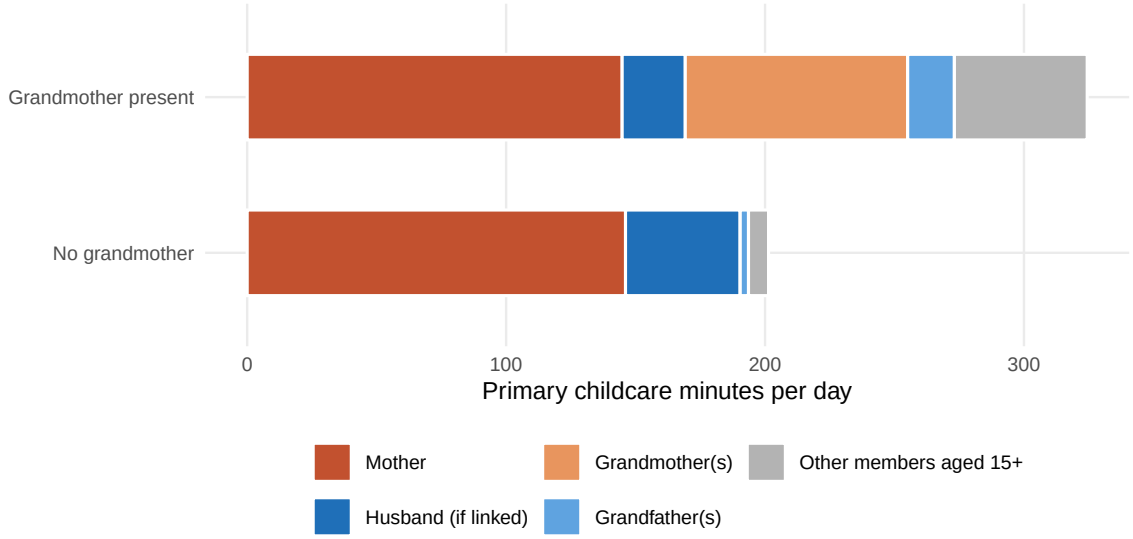


Figure 1: Primary childcare minutes per day by provider, with and without a co-resident grandmother. Values and standard errors in Table 3.

Table 3 and Figure 1 decompose the childcare provided by household members. In households with a grandmother, grandmothers provide 85.9 minutes (SE 1.1) of childcare a day. They also report 148.7 minutes (SE 1.6) of domestic work and 98.9 minutes (SE 2.1) of employment, which shows what else their day contains. The sum of the mother's and the grandmother's childcare cannot be compared with the mother's alone, because adding a non-negative term mechanically raises the total. The informative comparison is of all adult members' childcare across household types, which conditions on the number of children of each age. Summed over all members aged 15 and above, childcare is 88.6 minutes (SE 2.3) higher in households with a grandmother. Children in three-generation households thus receive more adult childcare time, not the same care more cheaply. Part of the

difference may reflect two adults recording care of the same child at the same time, and part the care of several grandchildren at once; the diary does not identify the recipient. Husbands' childcare, for those linked, differs by -14.1 minutes (SE 0.8) with a grandmother's presence. Grandmothers thus also substitute for fathers' modest contributions.

6.3 The older generation's trade-offs

Table 4: The older generation: time use and a child under six in the household

Outcome	Mean, women	Grandmothers	Grandfathers	Difference
Childcare (primary)	4.0	79.48*** (1.08)	29.76*** (0.75)	49.72*** (1.31)
Active childcare	3.4	70.43*** (1.03)	26.86*** (0.71)	43.57*** (1.25)
Passive childcare (minding)	0.5	7.65*** (0.42)	2.42*** (0.22)	5.23*** (0.47)
Unpaid domestic work	149.6	-16.57*** (1.98)	-1.22 (0.93)	-15.35*** (2.19)
Employment (incl. travel)	74.5	4.73* (2.47)	11.80*** (4.13)	-7.07 (4.81)
Employed (usual status, share)	0.22	0.02*** (0.01)	0.03*** (0.01)	-0.01 (0.01)
Free time	392.5	-53.25*** (2.44)	-33.54*** (3.14)	-19.71*** (3.98)
Sleep	595.8	-9.19*** (1.43)	-6.16*** (1.69)	-3.03 (2.21)

Notes: Members aged 45–80 in the generation above a married woman of childbearing age (26,062 women, 19,382 men). Coefficient (SE) on an indicator for a child under six in the household, separately for women and men; Difference: women minus men, with a standard error that ignores the covariance of the two estimates (conservative, since grandmothers and grandfathers often share a household). Mean: women without a child under six. Controls: age band, education, marital status, urban, consumption quintile, children 6–14, caste, religion, special care need, proxy report; state and day fixed effects. FSU-clustered standard errors. * p<0.10, ** p<0.05, *** p<0.01.

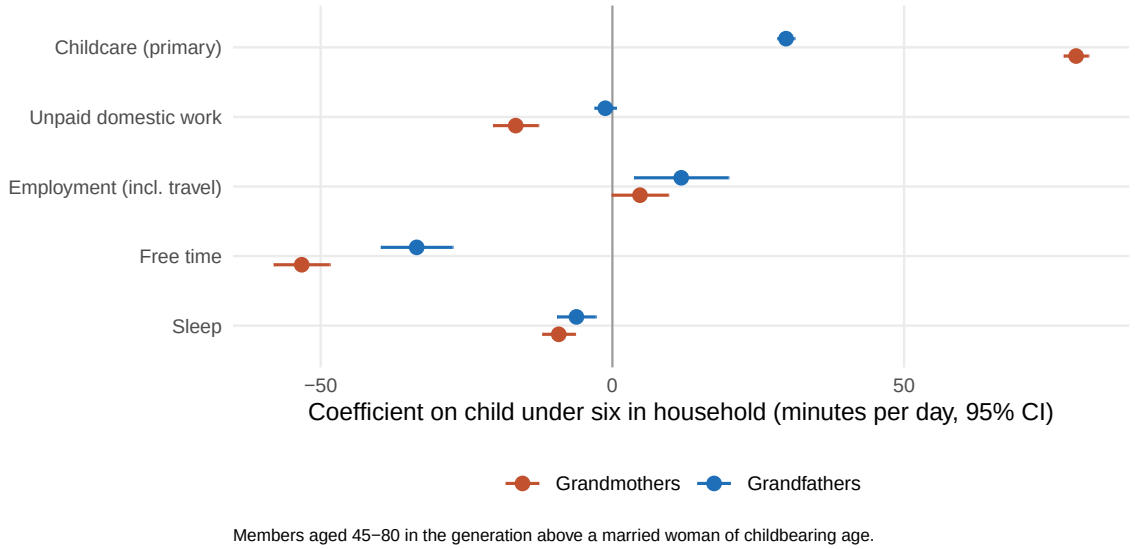


Figure 2: Older generation: minutes per day associated with a child under six in the household. Estimates in Table 4.

Table 4 and Figure 2 turn to the providers of care. Among older-generation women, a child under six in the household is associated with 79.5 minutes (SE 1.1) more primary childcare a day, against a mean of 4.0 minutes for women without a young grandchild. On the care side the third prediction of

the framework holds: grandmothers report more childcare when a young grandchild is present. The associations with grandmothers' domestic work (-16.6 minutes (SE 2.0)), paid work (4.7 minutes (SE 2.5)), employment status (1.6 percentage points (SE 0.6) higher), free time (-53.2 minutes (SE 2.4)) and sleep (-9.2 minutes (SE 1.4)) indicate where that time comes from. Grandmothers with a young grandchild report significantly less domestic work, free time and sleep, so the care they provide has a cost in their own time.

For older-generation men, the corresponding childcare association is 29.8 minutes (SE 0.7), and the women-minus-men difference is 49.7 minutes (SE 1.3). The care response of the older generation falls overwhelmingly on grandmothers, mirroring the gender division of care in the middle generation. Paid work of older men differs by 11.8 minutes (SE 4.1).

6.4 Which grandparents matter

Table 5: Mothers' childcare and employment minutes by type of co-resident grandparent

Comparison	Childcare	Employment
Grandmother under 60	3.06** (1.53)	-3.52* (2.03)
Grandmother 60 or older	-8.80*** (1.52)	3.64 (2.22)
Difference (under 60 minus 60+)	11.86*** (1.72)	-7.16*** (2.47)
Grandmother not employed	-5.24*** (1.37)	-2.75 (1.91)
Grandmother employed	5.82*** (1.93)	5.46* (2.79)
Difference (not employed minus employed)	-11.06*** (1.97)	-8.21*** (2.99)
Grandmother only	-8.88*** (1.63)	9.77*** (2.65)
Grandfather only	-5.76** (2.29)	-6.73** (3.09)
Both grandparents	1.12 (1.54)	-8.13*** (1.93)
Difference (grandmother only minus grandfather only)	-3.12 (2.52)	16.50*** (3.83)

Notes: Each block is one regression per outcome with mutually exclusive indicators; the omitted group is mothers with no co-resident grandmother (first two blocks) or no co-resident grandparent (third block). Age is that of the youngest co-resident grandmother; employment is usual principal activity status. Controls as in the main specification. Coefficient (SE); FSU-clustered. * p<0.10, ** p<0.05, *** p<0.01.

Table 5 examines the fourth prediction. Grandmother age is a proxy for health and stamina, which the survey does not measure directly for the older generation. Relative to mothers without a grandmother, the coefficient on mothers' childcare is 3.1 minutes (SE 1.5) for a grandmother under 60 and -8.8 minutes (SE 1.5) for a grandmother aged 60 or older; the difference is 11.9 minutes (SE 1.7). Older grandmothers are associated with a larger reduction, which does not fit a simple capacity story and may reflect that older grandmothers are less likely to work.

An employed grandmother has less time to give. The coefficient is -5.2 minutes (SE 1.4) for a grandmother who is not employed and 5.8 minutes (SE 1.9) for one who is, a difference of -11.1 minutes (SE 2.0). Substitution is concentrated among grandmothers who are not in paid work. Grandmothers' own employment is itself a choice, made jointly with the care they provide, so this comparison is descriptive.

The last block separates grandmothers from grandfathers. The coefficient on mothers' childcare is -8.9 minutes (SE 1.6) for a grandmother without a grandfather, -5.8 minutes (SE 2.3) for a grandfather without a grandmother, and 1.1 minutes (SE 1.5) for both together. The difference between grandmothers and grandfathers is not statistically significant; households with only a grandfather are few, so this comparison is imprecise. A pure selection story, in which three-generation households differ from nuclear ones in some fixed trait, would not predict that the characteristics of the older generation matter in this way.

6.5 Heterogeneity

Table 6: Heterogeneity: grandmother co-residence and mothers' childcare and employment minutes

Subsample	Childcare	Employment	N
Head-level mother (grandparents live in her household)	-13.21*** (2.07)	12.55*** (3.55)	32,742
Junior-level mother (lives in the older generation's household)	3.36 (2.62)	7.57** (3.24)	17,250
Urban	-6.78*** (2.48)	12.83*** (4.00)	16,812
Rural	-0.47 (1.48)	-5.88*** (1.84)	33,180
Mother below secondary education	-1.04 (1.73)	-6.91*** (2.66)	24,804
Mother secondary education or above	-3.52** (1.79)	5.05** (2.32)	25,188
One child under six	-0.10 (1.58)	1.30 (2.64)	26,024
Two or more children under six	-4.35** (1.86)	-2.67 (2.16)	23,968

Notes: Main specification, estimated separately by subsample. Coefficient (SE) on the grandmother indicator. Not estimable: fewer than 30 mothers with or without a co-resident grandmother in the subsample. * p<0.10, ** p<0.05, *** p<0.01.

Table 6 reports the estimates by subsample. For head-level mothers, whose parents or in-laws live in their household, the childcare association is -13.2 minutes (SE 2.1). For junior-level mothers, who live in their in-laws' or parents' household, it is 3.4 minutes (SE 2.6), estimated from only 2,054 junior-level mothers without a grandmother. The two arrangements are selected for different reasons, the first more often by the grandparents' need for support and the second by the patrilocal norm.

Across urban and rural households the childcare associations are -6.8 minutes (SE 2.5) and -0.5 minutes (SE 1.5). By the mother's education they are -1.0 minutes (SE 1.7) below secondary and -3.5 minutes (SE 1.8) at secondary or above. More educated mothers have better labour-market options, so if grandmothers' help reaches paid work anywhere it should be among them; the corresponding employment associations are -6.9 minutes (SE 2.7) and 5.1 minutes (SE 2.3). With one child under six the childcare association is -0.1 minutes (SE 1.6), and with two or more -4.3 minutes (SE 1.9).

6.6 A placebo: mothers of school-age children

If grandmothers substitute for the care of young children, the association with mothers' childcare should be weaker where the youngest child is of school age and needs less supervision. If instead the main estimate reflects fixed differences between joint and nuclear households, it should be similar for both groups of mothers. Table 7 reports the comparison for 19,829 mothers whose youngest

Table 7: Placebo: grandmother co-residence for mothers whose youngest child is aged 6 to 14

Outcome	Child under six	Youngest child 6–14	Difference
Childcare (primary)	-2.09* (1.27)	-0.34 (0.98)	-1.75 (1.60)
Active childcare	-2.17* (1.24)	0.44 (0.94)	-2.61* (1.55)
Passive childcare (minding)	-0.05 (0.26)	-0.38*** (0.14)	0.33 (0.30)
Unpaid domestic work	1.14 (1.70)	8.81*** (2.96)	-7.68** (3.41)
Employment (incl. travel)	-0.41 (1.73)	-0.06 (3.78)	-0.35 (4.16)

Notes: Coefficient (SE) on the grandmother indicator. Column 1: analysis sample (49,992 mothers). Column 2: mothers defined in the same way with no child under six in the household and an identified youngest own child aged 6 to 14 (19,829 mothers). Difference: column 1 minus column 2, with a standard error that ignores the covariance of the two estimates. Controls as in the main specification. * p<0.10, ** p<0.05, *** p<0.01.

identified own child is aged 6 to 14 and who have no child under six in the household. The childcare association is -0.3 minutes (SE 1.0) in this group, against -2.1 minutes (SE 1.3) in the analysis sample, and the difference between the two is -1.7 minutes (SE 1.6). Since the main childcare estimate is itself not significantly negative, the placebo mainly confirms the absence of a substitution pattern that is specific to young children. The association with domestic work, which the constraint channel predicts should not depend on the child's age, is 8.8 minutes (SE 3.0) in the placebo group and 1.1 minutes (SE 1.7) in the analysis sample.

7 Robustness

Table 8: Robustness: mothers' primary childcare minutes

Check	Estimate	SE
Baseline	-2.09*	1.27
Only married woman aged 18–49 in the household	-3.18**	1.36
Head-level mothers only (head or spouse of head)	-13.21***	2.07
Mothers with a linked husband only	-1.79	1.36
Self-reported diaries only	-3.34**	1.46
Controlling for other women aged 15+	-0.93	1.27
Controlling for household size	8.25***	1.60
Controlling for mother's generation (junior)	-6.75***	1.62
Unweighted	-1.81	1.12
Any grandparent instead of grandmother	-3.26***	1.26
Extensive margin: any primary childcare (share)	-0.018***	0.004
Oster bias-adjusted, childcare (delta = 1, Rmax = 1.3 R)	-2.33	
Oster bias-adjusted, employment (delta = 1, Rmax = 1.3 R)	-1.00	

Notes: Coefficient on the grandmother indicator (or any grandparent) in the main specification, unless stated. The extensive-margin row is a linear probability model. The Oster rows use the short regression without controls or fixed effects on the same sample. * p<0.10, ** p<0.05, *** p<0.01.

Table 8 collects robustness checks for the childcare estimate. Restricting to households with a single married woman aged 18 to 49, where the child under six is most likely hers, gives -3.2 minutes (SE

1.4). Restricting to head-level mothers, for whom the comparison is between nuclear and extended households of the same type, gives -13.2 minutes (SE 2.1). Requiring a linked husband gives -1.8 minutes (SE 1.4), and dropping proxy-reported diaries gives -3.3 minutes (SE 1.5). Adding the number of other adult women, who may also provide care, gives -0.9 minutes (SE 1.3); adding household size gives 8.2 minutes (SE 1.6); and adding an indicator for the mother's generation gives -6.8 minutes (SE 1.6). The unweighted estimate is -1.8 minutes (SE 1.1), and broadening the treatment to any grandparent gives -3.3 minutes (SE 1.3). On the extensive margin, a grandmother is associated with a 1.8 percentage points (SE 0.4) lower probability that the mother reports any primary childcare. The estimates range from -13.2 to 8.2 minutes and do not all share a sign, so the main childcare estimate should be read as fragile.

The bias-adjusted coefficient of Oster (2019), which assumes that unobservables are as important as observables and that the maximum attainable R^2 is 1.3 times that of the full model, is -2.33 for childcare and -1.00 for employment.

8 Discussion

8.1 Interpretation

The diaries of both generations give a direct view of how three-generation households in India divide the care of young children. Co-resident grandmothers are not associated with significantly less childcare by mothers. Grandmothers report more childcare themselves when a young grandchild is present. Children in three-generation households receive more adult childcare time in total. Mothers' paid work is not significantly higher with a grandmother present.

Three readings are consistent with this pattern, and the evidence allows a partial ranking. The first is that grandmothers' care is a complement to mothers' care rather than a substitute: the young child receives attention from two adults, and the mother's time with the child is protected by norms of what a mother should do. The second is the constraint channel: a larger household and a senior woman's authority raise the mother's domestic load and the wedge on work outside the home, so whatever childcare time is released is absorbed. The third is selection: mothers who choose, or are placed in, three-generation households differ in ways that also shape their time. The tests of Sections 6 and 7 speak to the third reading. Associations that vary with the sex, age and employment of the older generation, that are concentrated among mothers of young children, and that survive coefficient-stability bounds are harder to attribute to fixed household traits; associations that do not are easier to.

The comparison with the literature is instructive. Studies in the United States, Europe, Japan and China find that grandparents raise maternal employment (Compton and Pollak 2014; Dimova and Wolff 2011; Bratti et al. 2018; Sasaki 2002; Maurer-Fazio et al. 2011). This paper finds no such association in India, although grandmothers provide 37% of the combined childcare of the two women. The contrast is consistent with the view that the binding constraint on young Indian

mothers' employment is not, or not only, the availability of childcare, and with the evidence that household structure and norms restrict married women's work in India (Dhanaraj and Mahambare 2019; Anukriti et al. 2020; Jayachandran 2021). The Norwegian finding that formal childcare mainly crowded out informal care (Havnes and Mogstad 2011) provides a useful parallel: when one form of care replaces another, the effect on mothers' work depends on what else constrains it.

8.2 Policy

India's main public childcare instrument, the *anganwadi*, was designed around child nutrition and pre-school education rather than full-day care for working mothers. An implicit assumption in policy debate is that the extended family fills the gap. The results speak to that assumption in two ways. First, a co-resident grandmother is not associated with more paid work by mothers, so family care does not substitute for formal childcare in its labour-market effect, even where it is available. Expanding formal childcare may matter for mothers in extended households as well as in nuclear ones, although its effect will depend on whether the constraints of the extended household also bind on the use of formal care. Second, family care is not free: it is provided by older women whose own time responds to the presence of a young grandchild. As India's population ages (UNFPA and International Institute for Population Sciences 2023; International Institute for Population Sciences et al. 2020), the supply of grandmother care will compete with the older generation's own work and care needs, and its implicit cost belongs in any assessment of public childcare.

8.3 Limitations

The limitations are those of a single cross-section of diaries. Living arrangements are chosen, and the comparisons across arrangement types, the gradients by grandparent type, the placebo and the coefficient-stability bounds narrow but do not eliminate the scope for selection. Candidate instruments used elsewhere, such as the survival of grandparents or the husband's birth order, are not available in the survey. One diary day per person measures a day, not a routine (Frazis and Stewart 2012), which adds noise to individual outcomes but does not bias population means. Children under six are counted as a residual from household size and are assigned to a mother by a generational rule; I show that the main estimate is robust to restricting to households where the assignment is unambiguous, but misassignment in households with several junior couples cannot be ruled out. The roster code for parents of the head does not distinguish the head's own parents from the spouse's parents, so for head-level mothers I cannot separate the maternal from the paternal grandmother. Grandmother age proxies for health only imperfectly. Finally, the survey does not record which child a care episode is for. In households with several young children, or with children of more than one couple, the grandmother's recorded care may serve children other than the mother's. Future rounds that record the recipient of care, and household surveys that follow the same families over the birth of a grandchild, would allow sharper tests.

9 Conclusion

Diaries for every member of the household allow a direct test of an assumption that runs through discussion of the care economy in India: that co-resident grandmothers lighten the load of mothers of young children. Identifying the generation above each mother correctly matters for the test, because in patrilocal households the young mother lives in her parents-in-law's household and, without care, the head's wife would be counted as a mother rather than as the grandmother she is. With that identification, co-resident grandmothers are not associated with less childcare by mothers or with more paid work by them. Grandmothers themselves provide 37% of the combined childcare of the two women, and their own childcare is higher when a young grandchild is present. Family care in India is therefore best understood not as a free substitute for formal childcare, but as a transfer of time between women of two generations whose terms are set by the structure of the household.

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Appendix

Table 9: Construction and validation of the person-day panel

Check	Value
Activity records	10,211,478
Person-days (diaries)	533,719
Households	139,487
First Stage Units	9,969
Diaries summing to 1,440 minutes	100.0%
Records flagged minor (secondary)	17.5%
Persons with varying weight	0.00%
Invalid activity codes	None
Married persons with linked spouse	88.5%
Under-6 count method	direct count (file contains members under 6)

Notes: All activity records are used; no row sampling.

Table 10: Sample construction

Step	N
Married women 18–49 at head or junior level, child under six in household	52,181
Excluded: head-level women living with a junior married woman aged 15–49	2,189
Analysis sample (mothers of children under six)	49,992
with a co-resident grandmother	18,028
junior-level mothers without a grandmother	2,054
head-level mothers without a grandmother	29,910
Placebo sample (youngest own child 6–14, no child under six)	19,829
Older generation: women aged 45–80	26,062
Older generation: men aged 45–80	19,382

Notes: Unweighted counts of person-day diaries. Head level: head or spouse of head. Junior level: married child of head or spouse of a married child.